The skeleton of an acyclic cluster variety

Cluster skeleton geometry: formalization reference

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This reference replaces the previous project PDF and follows the supplied "Skeleton in the acyclic case" proof. Its scope is the smooth Liouville primitive, complete flow, and the explicit dynamical skeleton. The mathematical proof below is the informal input to Lean. Its presence here does not certify that the corresponding Lean theorem has been completed. The existing skew-symmetric unimodular target is the specialization $\widetilde{D} = I$; the weighted notation is retained from the supplied proof.

1 Skeleton in the acyclic case

Let M and F be the mutable and frozen index sets, and put $n = |M| + |F|$. Fix an integral completion $E \in GL_n(\mathbb{Z})$ whose mutable columns are $\widetilde{B}$. Fix a positive rational diagonal matrix $\widetilde{D} = \mathrm{diag}(d_j)$ such that $K = E\widetilde{D}$ is skew-symmetric. The skew-symmetric case is $\widetilde{D} = I_n$ and $K = E$. The mutable block $B = E_{M,M}$ is acyclic: there is no directed cycle with all successive matrix entries positive. Frozen coordinates remain invertible variables. The unimodular completion hypothesis is retained throughout.

Write $c_i = Ee_i$ and $z^{c_i} = \prod_j z_j^{E_{ji}}$. On the initial torus put

$$\Omega = \frac{1}{2}\sum_{j,l} K_{jl}\, d\log z_j \wedge d\log z_l, \qquad \lambda_{\mathrm{tor}} = u^{\mathsf{T}} K\, d\theta.$$

Here Ω denotes its smooth holomorphic symplectic extension to the exchange variety. The construction and nondegeneracy of that extension, and the smooth atlas on the actual exchange variety, are separate Lean construction obligations; they must not be hidden in the final theorem's hypotheses. The local node charts below give the coordinates for them. For a complete Liouville field, define its dynamical skeleton as the set of points with relatively compact forward orbit. Our sign convention is $\iota_Z d\lambda = \lambda$.

A subset $I \subset M$ is *independent* if $B_{ij} = 0$ for all $i, j \in I$.

In the acyclic case, $\mathcal{A}(\widetilde{B})$ can be presented by generators and relations. For $i \in M$ put

$$P_i^{\pm} = \prod_{j \in M \sqcup F} z_j^{[\pm \widetilde{B}_{ji}]_+}, \qquad t_i = P_i^+ / P_i^- = z^{c_i}.$$

By [1, Corollary 1.21],

$$\mathcal{A}(\widetilde{B}) = \frac{\mathbb{C}[z_i, z_i' \ (i \in M),\ z_j^{\pm 1} \ (j \in F)]}{(z_i z_i' - P_i^+ - P_i^-)_{i \in M}}. \tag{1}$$

Lemma 1. *We have the following.*

(i) *At every point $p \in \mathcal{A}(\widetilde{B})$, the set $\{i \in M \mid z_i(p) = 0\}$ is independent.*

(ii) *Let I be independent, and let $U_I = \{z_j \neq 0 \ (j \in M \setminus I)\} \subset \mathcal{A}(\widetilde{B})$. Then $U_I \cong \mathcal{N}^I \times (\mathbb{C}^*)^{m+f-2|I|}$, where the ith factor $\mathcal{N}$ has coordinates*

$$(x, y, t) = (z_i,\ z_i'/P_i^-,\ t_i) \qquad (i \in I).$$

(iii) *The open sets U_I cover $\mathcal{A}(\widetilde{B})$.*

Proof. (i) Fix a point p, let $Z = \{i \in M : z_i(p) = 0\}$, and suppose that $B_{Z,Z} \neq 0$. Since B is acyclic, we can take a maximal path $j_1, \dots, j_r$ in Z with $r \geq 2$ and $B_{j_s j_{s+1}} > 0$. Put $i = j_r$. By maximality, $B_{ji} \geq 0$ for all $j \in Z$, and $B_{j_{r-1} i} > 0$. So $P_i^+(p) = 0$ since $z_{j_{r-1}}(p) = 0$. Every mutable j in P_i^- has $B_{ji} < 0$, so $j \notin Z$; the frozen z_j are units. So $P_i^-(p) \neq 0$. This contradicts the relation $z_i z_i' = P_i^+ + P_i^-$.

(ii) On U_I, the z_j with $j \in (M \sqcup F) \setminus I$ are units. For $i \in I$, the monomials $P_i^\pm$ involve only these z_j. For $j \in M \setminus I$, we have $z_j' = (P_j^+ + P_j^-)/z_j$. So (1) gives

$$\mathcal{O}(U_I) = \mathbb{C}[z_j^{\pm 1} \ (j \in M \sqcup F \setminus I)]\,[z_i, y_i \ (i \in I)]/(z_i y_i - 1 - t_i)_{i \in I}, \qquad y_i = z_i'/P_i^-.$$

The vectors c_i, $i \in I$, are columns of $E \in GL_{m+f}(\mathbb{Z})$ and lie in $\mathbb{Z}^{(M \sqcup F) \setminus I}$. So they extend to a basis of $\mathbb{Z}^{(M \sqcup F) \setminus I}$, which implies $\mathbb{C}[z_j^{\pm 1} \ (j \notin I)] = \mathbb{C}[t_i^{\pm 1} \ (i \in I)] \otimes \mathcal{O}\big((\mathbb{C}^*)^{m+f-2|I|}\big)$. This proves (ii).

(iii) For any point $p \in \mathcal{A}(\widetilde{B})$, take $I = \{i \in M \mid z_i(p) = 0\}$. Then $p \in U_I$. □

Cutoffs and the local node

The node is $\mathcal{N} = \{(x, y, t) \in \mathbb{C}^2 \times \mathbb{C}^* : xy = 1 + t\}$. On $x \neq 0$ write $x = e^{u+i\theta}$ and $t = e^{v+i\eta}$. Its holomorphic symplectic form is $d\log t \wedge d\log x = dy \wedge dx/(xy - 1)$. Choose $L > a > 0$ and smooth functions satisfying

$$\begin{gathered}
h \leq 0, \quad 0 \leq h' \leq 1, \quad u - h(u) < 0 \ (u < 0), \\
h(u) = u + a \ (u \leq -L), \qquad h = 0 \text{ near } [0, \infty), \\
\rho \in C_c^\infty(\mathbb{R}, [0, 1]), \qquad \rho = 1 \text{ near } 0, \\
\beta \in C^\infty(\mathbb{R}/2\pi\mathbb{Z}), \quad \operatorname{supp}\beta \subset (\pi - \varepsilon, \pi + \varepsilon), \\
\beta(\eta) = \eta - \pi \text{ near } \pi, \quad (\eta - \pi)\beta(\eta) \geq 0, \quad \beta' \leq 1, \qquad 0 < \varepsilon < \pi.
\end{gathered}$$

The angular difference is the lift in $(-\varepsilon, \varepsilon)$ on this arc. The node primitive is $v\,d\theta - u\,d\eta + d(h\rho\beta)$. These cutoff properties, including existence, are part of the formalization.

Theorem 1. *There is a Liouville form λ_{ac} of $\Im\Omega$ on $\mathcal{A}(\widetilde{B})$ such that*

$$\begin{gathered}
\Lambda = \operatorname{Skel}(Z_{\lambda_{\mathrm{ac}}}) = \bigcup_{I \subset M \ \textit{independent}} \Lambda_I, \\
\Lambda_I = \{z_i' = 0,\ |z_i| \leq 1 \ (i \in I), \quad |z_j| = 1 \ (j \in (M \sqcup F) \setminus I)\}.
\end{gathered} \tag{2}$$

Proof. Use the constants $L > a > 0$ and the cutoffs h, ρ, β specified above. Let $\lambda_{\mathrm{tor}} = u^{\mathsf{T}} K \mathrm{d}\theta$. Fix $\delta_0 > 0$ with $h = 0$ on $(-\delta_0, \infty)$. Choose a smooth nonincreasing $\vartheta\colon \mathbb{R} \to [0, 1]$ with $\vartheta = 1$ on $(-\infty, -1]$ and $\vartheta = 0$ on $[0, \infty)$. On the initial torus $(\mathbb{C}^*)^{m+f}$ write

$$u_j = \log|z_j|, \qquad \theta_j = \arg z_j, \qquad v_i = \log|t_i| = c_i(u), \qquad \eta_i = \arg t_i = c_i(\theta),$$

and set

$$\chi_i \coloneqq \prod_{j \in M\colon B_{ji} \neq 0} \vartheta(u_i - u_j), \qquad H \coloneqq \sum_{i \in M} d_i\, h(u_i)\, \chi_i\, \rho(v_i)\, \beta(\eta_i), \qquad \lambda_{\mathrm{ac}} \coloneqq \lambda_{\mathrm{tor}} + \mathrm{d}H.$$

An empty product is 1. Note that if $B_{ij} \neq 0$, then $\chi_i \chi_j = 0$.
Let $I \subset M$ be independent. For $i, i' \in I$, $K_{ii'} = d_{i'} B_{ii'} = 0$. For $i \in M$,

$$\sum_j K_{ji}\, u_j = d_i \sum_j E_{ji} u_j = d_i v_i, \qquad \sum_l K_{il}\, \mathrm{d}\theta_l = -\sum_l K_{li}\, \mathrm{d}\theta_l = -d_i\, \mathrm{d}\eta_i.$$

The terms of λ_{tor} with $l \in I$ give $\sum_{i\in I} d_i v_i\, \mathrm{d}\theta_i$. The terms with $j \in I$ and $l \notin I$ give $-\sum_{i\in I} d_i u_i\, \mathrm{d}\eta_i$, since $K_{il} = 0$ for $i, l \in I$. Hence

$$\lambda_{\mathrm{tor}} = \sum_{i\in I} d_i (v_i\, \mathrm{d}\theta_i - u_i\, \mathrm{d}\eta_i) + \sum_{j,l\in (M\sqcup F)\setminus I} K_{jl}\, u_j\, \mathrm{d}\theta_l. \tag{3}$$

We first show that λ_{ac} is globally defined. Let $q \in \mathcal{A}(\widetilde{B}) \setminus (\mathbb{C}^*)^{m+f}$, and let $I = \{i \in M \mid z_i(q) = 0\}$. Then $I \neq \emptyset$, I is independent by Lemma 1(i), and $q \in U_I$. Let $i \in I$. The monomials $P_i^{\pm}$ only involve z_j with $j \notin I$, so they are units on U_I. Hence $t_i = P_i^+ / P_i^-$ is a unit on U_I, and $0 = z_i(q) z_i'(q) = P_i^-(q)(1 + t_i(q))$ gives $t_i(q) = -1$. So, as $p \to q$ with $p \in (\mathbb{C}^*)^{m+f}$,

$$u_i(p) \to -\infty, \qquad v_i(p) \to 0, \qquad \eta_i(p) \to \pi, \qquad u_j(p) \to \log|z_j(q)| \in \mathbb{R} \quad \big(j \in (M \sqcup F) \setminus I\big).$$

Choose a neighborhood $W \subset U_I$ of q such that, for $i \in I$ and $j \in M$ with $B_{ji} \neq 0$,

$$\rho(v_i) = 1, \quad \beta(\eta_i) = \eta_i - \pi \quad \text{on } W, \qquad u_i \leq -L, \quad u_i \leq u_j - 1 \quad \text{on } W \cap (\mathbb{C}^*)^{m+f}.$$

Then, on $W \cap (\mathbb{C}^*)^{m+f}$, $h(u_i) = u_i + a$ and $\chi_i = 1$. Also $\chi_j = 0$ there, because χ_j has the factor $\vartheta(u_j - u_i)$ and $u_j - u_i \geq 1$. So, on $W \cap (\mathbb{C}^*)^{m+f}$,

$$H = \sum_{i\in I} d_i (u_i + a)(\eta_i - \pi) + H', \tag{4}$$

where H' is the sum of the summands of H with index $k \in M \setminus I$ such that $B_{ik} = 0$ for all $i \in I$. For such k, the functions u_k, χ_k, $v_k = c_k(u)$ and $\eta_k = c_k(\theta)$ only involve z_j with $j \notin I$, by the support of c_k. These z_j are units on U_I. So H' does not depend on u_i and θ_i, $i \in I$.
By (3) and (4), on $W \cap (\mathbb{C}^*)^{m+f}$,

$$\lambda_{\mathrm{ac}} = \sum_{j,l\notin I} K_{jl}\, u_j\, \mathrm{d}\theta_l + \mathrm{d}H' + \sum_{i\in I} d_i (v_i\, \mathrm{d}\theta_i + (\eta_i - \pi)\, \mathrm{d}u_i + a\, \mathrm{d}\eta_i).$$

Here we used $v_i\, \mathrm{d}\theta_i - u_i\, \mathrm{d}\eta_i + \mathrm{d}\big((u_i + a)(\eta_i - \pi)\big) = v_i\, \mathrm{d}\theta_i + (\eta_i - \pi)\, \mathrm{d}u_i + a\, \mathrm{d}\eta_i$. The first two terms are smooth on W. On W, $1 - z_i y_i = -t_i$ and $\mathrm{Log}(-t_i) = v_i + \mathrm{i}(\eta_i - \pi)$. So, by the holomorphic logarithm near 1,

$$v_i\, \mathrm{d}\theta_i + (\eta_i - \pi)\, \mathrm{d}u_i + a\, \mathrm{d}\eta_i = \Im\Big(\frac{\mathrm{Log}(1 - z_i y_i)}{z_i}\, \mathrm{d}z_i\Big) + a\, \mathrm{d}\arg(1 - z_i y_i).$$

The right side is smooth on W, because $\mathrm{Log}(1 - z_i y_i)/z_i$ is holomorphic on W, with value $-y_i$ on $\{z_i = 0\}$. Hence λ_{ac} extends smoothly over W, hence also extends to $\mathcal{A}(\widetilde{B})$.
Now we consider the Liouville field $Z = Z_{\lambda_{\mathrm{ac}}}$, given by $\iota_Z \Im\Omega = \lambda_{\mathrm{ac}}$. A dot denotes the derivative along Z. On the initial torus, $\Im\Omega = \sum_{j,l} K_{jl}\, \mathrm{d}u_j \wedge \mathrm{d}\theta_l$. Comparing the coefficients of $\mathrm{d}\theta$ and of $\mathrm{d}u$ in $\iota_Z \Im\Omega = \lambda_{\mathrm{tor}} + \mathrm{d}H$ gives

$$K^{\mathsf{T}} \dot{u} = K^{\mathsf{T}} u + \partial_\theta H, \qquad -K\dot{\theta} = \partial_u H.$$

The function H depends on θ only through $\eta_i = c_i(\theta)$, and $d_i c_i = K e_i$. So

$$\partial_\theta H = \sum_{i\in M} d_i\, h(u_i) \chi_i \rho(v_i) \beta'(\eta_i)\, c_i = K \sum_{i\in M} h(u_i) \chi_i \rho(v_i) \beta'(\eta_i)\, e_i.$$

Since $K^{\mathsf{T}} = -K$, this gives

$$\begin{aligned} \dot{u}_j &= u_j && (j \in F), \\ \dot{u}_i &= u_i - h(u_i)\,\chi_i\,\rho(v_i)\,\beta'(\eta_i) && (i \in M), \\ \dot{\theta} &= -K^{-1}\partial_u H. \end{aligned} \tag{5}$$

For $i \in M$, $c_i(K^{-1}x) = -x_i/d_i$, because $d_i c_i^{\mathsf{T}} = e_i^{\mathsf{T}} K^{\mathsf{T}} = -e_i^{\mathsf{T}} K$. Also $c_i(e_k) = E_{ki} = B_{ki}$ for $k \in M$. So applying c_i to (5) gives

$$\dot{\eta}_i = \frac{1}{d_i}\,\partial_{u_i} H, \qquad \dot{v}_i = v_i - \sum_{k\in M} B_{ki}\,h(u_k)\,\chi_k\,\rho(v_k)\,\beta'(\eta_k) \qquad (i \in M). \tag{6}$$

We now bound $\dot{u}$. Since $h \le 0$, $0 \le \chi_i\rho \le 1$ and $\beta' \le 1$, we have $h\chi_i\rho\beta' \ge h\chi_i\rho \ge h$. With $h = 0$ on $(-\delta_0, \infty)$, (5) gives, for $i \in M$,

$$\dot{u}_i \le u_i - h(u_i) \quad \text{on } \{u_i < 0\}, \qquad \dot{u}_i = u_i \quad \text{on } \{u_i > -\delta_0\}. \tag{7}$$

Recall that $u - h(u) < 0$ for $u < 0$, and $u - h(u) = -a$ for $u \le -L$. Also $|h(u)| \le |u| + L$, since $h(u) = u + a$ for $u \le -L$, and $a - L \le h \le 0$ on $[-L, \infty)$. So (5) gives $|\dot{u}_j| \le C(1 + |u_j|)$ for all $j \in M \sqcup F$, with $C = (1 + \max|\beta'|)(1 + L)$. By density, (7) holds on $\{z_i \ne 0\}$, and the bound holds on $\{z_j \ne 0\}$.

Next we compute Z near $\{z_i = 0\}$. Let q, I and W be as above, and let $i \in I$. On $W \cap (\mathbb{C}^*)^{m+f}$, $h(u_i) = u_i + a$, $\chi_i = \rho(v_i) = \beta'(\eta_i) = 1$, and $\chi_k = 0$ for $B_{ki} \ne 0$. Also $\partial_{u_i} H = d_i(\eta_i - \pi)$ there, by (4). So (5) and (6) give

$$\dot{u}_i = -a, \qquad \dot{v}_i = v_i, \qquad \dot{\eta}_i = \eta_i - \pi \qquad \text{on } W \cap (\mathbb{C}^*)^{m+f}.$$

Since $\operatorname{Log}(-t_i) = v_i + \mathrm{i}(\eta_i - \pi)$, by density

$$Z|z_i|^2 = -2a|z_i|^2, \qquad \dot{t}_i = t_i \operatorname{Log}(-t_i) \qquad \text{on } W. \tag{8}$$

Every point of $\{z_i = 0\}$ is such a q, so these sets W cover $\{z_i = 0\}$. Along an orbit near $\{z_i = 0\}$, $|z_i|^2$ solves $\frac{\mathrm{d}}{\mathrm{d}\tau}|z_i|^2 = -2a|z_i|^2$. By uniqueness of solutions, $\{z_i = 0\}$ and its complement are invariant under the flow of Z.

On $\{z_i = 0\}$, the z_j with $B_{ji} \ne 0$ are nonzero, by Lemma 1(i), so $P_i^- \ne 0$. So $y_i = z_i'/P_i^-$ is defined near $\{z_i = 0\}$, and $y_i = (1 + t_i)/z_i$ where $z_i \ne 0$. Put $\phi(t) = t\operatorname{Log}(-t)/(1 + t)$. It is holomorphic near $t = -1$, with $\phi(-1) = 1$. On $W \cap \{z_i y_i \ne 0\}$, (8) gives

$$Z \log|y_i| = \Re\frac{\dot{t}_i}{1 + t_i} - Z\log|z_i| = \Re\phi(t_i) + a.$$

By density, $Z|y_i|^2 = 2(\Re\phi(t_i) + a)|y_i|^2$ on W. On $W \cap \{z_i = 0\}$, $t_i = -1$, so

$$Z|y_i|^2 = 2(1 + a)|y_i|^2 \qquad \text{on } \{z_i = 0\}. \tag{9}$$

Hence $|y_i(\phi_Z^\tau(p))| = e^{(1+a)\tau}|y_i(p)|$ for $p \in \{z_i = 0\}$ and all τ.

Next we show that Z is complete. Let $\gamma\colon (\tau_-, \tau_+) \to \mathcal{A}(\widetilde{B})$ be a maximal integral curve of Z, and let $I_0 = \{i \in M \mid z_i(\gamma(0)) = 0\}$. By invariance, $z_i = 0$ on γ for $i \in I_0$, and $z_j \ne 0$ on γ for $j \in (M \sqcup F) \setminus I_0$. Let $T > 0$. For $\tau \in (\tau_-, \tau_+)$ with $|\tau| \le T$, the bound $|\dot{u}_j| \le C(1 + |u_j|)$ and (9) give

$$|u_j(\gamma(\tau))| \le (1 + |u_j(\gamma(0))|)e^{CT} \quad (j \notin I_0), \qquad |y_i(\gamma(\tau))| \le e^{(1+a)T}|y_i(\gamma(0))| \quad (i \in I_0).$$

So the coordinates $z_j^{\pm 1}$ $(j \notin I_0)$, $z_j' = (P_j^+ + P_j^-)/z_j$ $(j \in M \setminus I_0)$ and $z_i' = P_i^- y_i$ $(i \in I_0)$ are bounded on these $\gamma(\tau)$. By (1), $\mathcal{A}(\widetilde{B})$ is closed in $\mathbb{C}^{2m} \times (\mathbb{C}^*)^f$, so these $\gamma(\tau)$ lie in a compact

set. Hence $\tau_\pm = \pm\infty$, since a maximal integral curve with $\tau_+ < \infty$ or $\tau_- > -\infty$ leaves every compact set.

We now prove (2).

(i) $\mathrm{Skel}(Z) \subset \cup_I \Lambda_I$. Let $p \in \mathrm{Skel}(Z)$, so that the closure of $\gamma(\tau) = \phi_Z^\tau(p)$, $\tau \geq 0$, is compact. Put

$$I_0 = \{i \in M \mid z_i(p) = 0\}, \qquad I = I_0 \cup \{i \in M \mid z_i(p) \neq 0,\ u_i(p) < 0\}.$$

We show that $p \in \Lambda_I$.

We first determine u along γ. For $j \in F$, $u_j(\gamma(\tau)) = e^\tau u_j(p)$ is bounded, so $u_j(p) = 0$. Let $i \in M \setminus I_0$. If $u_i(p) > 0$, then $u_i(\gamma(\tau)) = e^\tau u_i(p)$ by (7), which is unbounded. If $u_i(p) = 0$, then $u_i = 0$ on γ, by (7) and uniqueness for $\dot{u}_i = u_i$. If $u_i(p) < 0$, then u_i decreases along γ, and (7) gives

$$\dot{u}_i \leq \sup_{u \leq u_i(p)} \big(u - h(u)\big) < 0 \qquad \text{on } \gamma.$$

The supremum is negative because $u - h(u) < 0$ for $u < 0$ and $u - h(u) = -a$ for $u \leq -L$. So $u_i(\gamma(\tau)) \to -\infty$. In summary,

$$u_j = 0 \text{ on } \gamma \quad (j \in (M \sqcup F) \setminus I), \qquad u_i(\gamma(\tau)) \to -\infty \quad (i \in I \setminus I_0), \qquad z_i = 0 \text{ on } \gamma \quad (i \in I_0).$$

Let q be a limit point of γ. Then $\{i \in M \mid z_i(q) = 0\} = I$, so I is independent by Lemma 1(i). Also $|z_j| = 1$ on γ and at q for $j \in M \setminus I$. So the closure of γ is a compact subset of U_I. On U_I, $z_i' = P_i^- y_i$ with P_i^- a unit, for $i \in I$. Since $|z_i(p)| \leq 1$ for $i \in I$ and $|z_j(p)| = 1$ for $j \notin I$, it remains to show that $y_i(p) = 0$ for $i \in I$.

Let $i \in I_0$. By (9), $|y_i(\gamma(\tau))| = e^{(1+a)\tau}|y_i(p)|$, which is bounded only if $y_i(p) = 0$.

Let $i \in I \setminus I_0$. We show that $t_i = -1$ on γ. The character c_i is supported on $F \sqcup \{j \in M \mid B_{ji} \neq 0\}$, which is disjoint from I. So $v_i = c_i(u) = 0$ on γ. Consider the open set

$$O = \{z_i \neq 0,\ |z_j| > e^{-\delta_0}\ (j \in M,\ B_{ji} \neq 0)\} \supset \gamma.$$

Besides the ith summand of H, only the summands with index j, $B_{ji} \neq 0$, depend on u_i, through χ_j and $v_j = c_j(u)$. These vanish on $O \cap (\mathbb{C}^*)^{m+f}$, because $h(u_j) = 0$ there. Also $\partial_{u_i} v_i = E_{ii} = 0$. So (6) gives, on $O \cap (\mathbb{C}^*)^{m+f}$, and then on O by density,

$$\dot{\eta}_i = \big(h'(u_i)\,\chi_i + h(u_i)\,\partial_{u_i}\chi_i\big)\,\rho(v_i)\,\beta(\eta_i).$$

The factor in front of $\beta(\eta_i)$ is ≥ 0, because $h' \geq 0$, $h \leq 0$, $\chi_i \geq 0$, $\rho \geq 0$, and $\partial_{u_i}\chi_i \leq 0$ as $\vartheta' \leq 0$. Since $(\eta - \pi)\beta(\eta) \geq 0$ and $\operatorname{supp}\beta \subset (\pi - \varepsilon, \pi + \varepsilon)$, the distance $|\eta_i - \pi|$ in $\mathbb{R}/2\pi\mathbb{Z}$ is nondecreasing along γ. On the other hand, $z_i(\gamma(\tau)) \to 0$, and y_i is bounded on γ. So $1 + t_i = z_i y_i \to 0$, and $|\eta_i - \pi| \to 0$. Hence $\eta_i = \pi$ on γ. With $v_i = 0$, this gives $t_i = -1$ on γ, so $y_i = (1 + t_i)/z_i = 0$ on γ. Thus $p \in \Lambda_I$, and $\mathrm{Skel}(Z) \subset \bigcup_I \Lambda_I$.

(ii) $\Lambda_I \subset \mathrm{Skel}(Z)$.

Let I be independent, and let $p \in \Lambda_I$. The set Λ_I is closed, and on it the coordinates $z_j^{\pm 1}$ $(j \notin I)$, z_i, $z_i' = 0$ $(i \in I)$ and $z_j' = (P_j^+ + P_j^-)/z_j$ $(j \in M \setminus I)$ are bounded. So Λ_I is compact, and it suffices to show $\phi_Z^\tau(p) \in \Lambda_I$ for $\tau \geq 0$. Let $\gamma(\tau) = \phi_Z^\tau(p)$. As above, $u_j = 0$ on γ for $j \notin I$. For $i \in I$, $z_i = 0$ on γ if $z_i(p) = 0$, and $u_i \leq 0$ on γ if $z_i(p) \neq 0$. It remains to show $y_i = 0$ on γ for $i \in I$. If $z_i(p) = 0$, this follows from (9) and $y_i(p) = 0$. If $z_i(p) \neq 0$, then $v_i = 0$ on γ, and η_i solves the equation for $\dot{\eta}_i$ above, with $\eta_i(p) = \pi$. Since $\beta(\pi) = 0$, uniqueness gives $\eta_i = \pi$ on γ. So $t_i = -1$ and $y_i = 0$ on γ. This proves (2). □

Formalization target and convention matching

All geometry is on the actual complex points of the exchange equations with their induced topology. Construct the smooth presentation, the extended symplectic form, the primitive, and a

jointly smooth flow defined for every real time. Define the coordinate cap and dynamical trapped set independently, and prove their equality for this same flow. The equivalent coordinate cap is

$$\{p : |z_f| = 1 \ (f \in F),\ |z_i| \le 1,\ (1 - |z_i|^2) z_i' = 0 \ (i \in M)\}.$$

For the converse equivalence, the indices with $|z_i| < 1$ have $z_i' = 0$; the zero-set argument applied to their exchange relations makes them independent (equivalently use the existing strict-cap-independence lemma). The existing Lean seed uses row exponents, whereas this reference uses column exponents. In the skew-symmetric specialization the two exchange monomials are interchanged, so their sum and the exchange variety agree. Exchange ratios are inverted. Check the sign of the logarithmic form and all angle-flow formulas explicitly when transporting the proof.

No principal-coefficient assumption is imposed. Unimodularity is retained; neither removal of acyclicity nor arbitrary singular matrices is asserted. The weighted statement requires updating the seed interface; the old skew-symmetric case must remain a specialization. Use native Mathlib geometry and the pinned FsFltz interfaces. A successful build of conditional lemmas does not establish the existence target. For each accepted theorem record its exact type and a transitive axiom audit; permit only `propext`, `Classical.choice`, and `Quot.sound`.

References

[1] A. Berenstein, S. Fomin, A. Zelevinsky, *Cluster algebras III: Upper bounds and double Bruhat cells*, Duke Math. J. 126 (2005), 1–52, Corollary 1.21.